

DATA VISUALIZATION USING POWER BI – SALES TARGET ANALYSIS

1. Introduction

Data visualization plays an important role in converting raw business data into meaningful insights. This project was developed using Microsoft Power BI to analyze sales performance, compare actual sales with sales targets, identify regional sales patterns, and evaluate business performance through interactive dashboards and DAX calculations.

The objective of this project is to transform raw sales data into visual reports that support business decision-making through calculations, analysis, and visualization techniques.

2. Project Objective

The objectives of this project are:

- Import and transform sales datasets into Power BI.
 - Establish relationships between multiple tables.
 - Create calculated columns and DAX measures.
 - Build interactive dashboards and visual reports.
 - Compare actual sales against targets.
 - Analyze sales performance across categories, locations, and time periods.
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3. Dataset Description

Three datasets were used in this project:

List of Orders

Contains order-related information including:

- Order ID
- Order Date

- Customer Name
- State
- City

Order Details

Contains transaction details:

- Order ID
- Category
- Sub-Category
- Amount
- Profit
- Quantity

Sales Target

Contains target values:

- Category
 - Segment
 - Sales Target
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4. Data Import and Preparation

Steps Performed:

1. Open Power BI Desktop.
 2. Import CSV files using Get Data.
 3. Open Power Query Editor.
 4. Verify and correct data types.
 5. Remove null values and unnecessary records.
 6. Load transformed data into Power BI.
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5. Data Modeling

Relationships were created between datasets to enable analysis.

Relationships:

- List of Orders[Order ID] → Order Details[Order ID]
- Sales Target[Category] → Order Details[Category]

Relationship Type:

One-to-Many

6. DAX Calculations

Calculated Columns

Category Type

Combines Category and Sub-Category.

DAX:

Category Type =
Category & " - " & Sub-Category

Revenue per Order

Calculates revenue for each order.

DAX:

Revenue per Order =
Amount × Quantity

Sales Category

Classifies orders based on average sales.

DAX:

```
Sales Category =  
IF(  
Amount > Average Amount,  
"Above Average",  
"Below Average"  
)
```

Measures

Order Count

Calculates total number of orders.

DAX:

```
Order Count =  
COUNT(Order ID)
```

Average Profit in Delhi

Calculates average profit generated in Delhi.

DAX:

```
Average Profit Delhi =  
CALCULATE(  
AVERAGE(Profit),  
City="Delhi"  
)
```

Year-to-Date Sales

Calculates cumulative sales.

DAX:

```
YTD Sales =  
TOTALYTD(  
SUM(Amount),
```

Order Date
)

7. Data Visualization

1. Sales Target Achievement by Category

Visual: Clustered Column Chart

Purpose:

Compare actual sales and target sales across categories.

Fields:

- Axis → Category
 - Values → Sales Amount
 - Secondary Values → Target
-

2. Max Profit Margin by Sub-Category

Visual: Donut Chart

Purpose:

Identify sub-categories generating maximum profit.

Fields:

- Legend → Sub-Category
 - Values → Maximum Profit
-

3. Monthly Sales Trend

Visual: Line Chart

Purpose:

Analyze sales growth over time.

Fields:

- Axis → Month
 - Values → Sales
-

4. Comparison of Profit and Quantity

Visual: Scatter Chart

Purpose:

Study relationship between sales quantity and profit.

Fields:

- X → Quantity
 - Y → Profit
 - Details → Sub-Category
-

5. Comparison of Total Sales and Target

Visual: Card + Multi-row Card

Purpose:

Display actual sales and targets for comparison.

Cards:

- Total Sales
- Total Target

Multi-row Card:

- Minimum Target by Segment
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6. Sales Performance Matrix

Visual: Matrix

Purpose:

Compare actual sales and target values across categories and months.

Fields:

Rows → Category

Columns → Month

Values:

- Actual Sales
- Target Sales

Conditional formatting was applied to highlight performance.

7. Geographic Sales Analysis

Visual: Map

Purpose:

Analyze total sales across cities.

Fields:

- Location → City
 - Size → Total Sales
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8. Sales Distribution by Sub-Category

Visual: Treemap

Purpose:

Understand contribution of each sub-category.

Fields:

- Group → Sub-Category
 - Values → Sales
-

9. Order Count Analysis by State

Visual: Funnel Chart

Purpose:

Visualize order distribution across states.

Fields:

- Group → State
 - Values → Order Count
-

8. Dashboard Formatting

Formatting activities performed:

- Applied report titles.
 - Enabled data labels.
 - Applied conditional formatting.
 - Improved readability using alignment and spacing.
 - Organized visuals into dashboard layout.
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9. Key Insights

- Sales performance differs across categories.
 - Certain cities generated higher sales than others.
 - Some categories exceeded targets while others underperformed.
 - Monthly trends revealed seasonal sales behavior.
 - Funnel and map visuals helped identify regional opportunities.
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10. Conclusion

This project demonstrates how Power BI can be used to transform raw business data into meaningful visual insights. Through DAX calculations, dashboard creation, and interactive

reporting, the project enabled sales monitoring, target comparison, and business performance analysis effectively.

The dashboard supports decision-making by presenting data in a simple, visual, and interactive manner.